

Hot-blade cutting of EPS foam for double-curved surfaces—numerical simulation and experiments

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Abstract In the present paper, experimental and numerical studies of a newly developed process of Hot-Blade Cutting used for free forming of double-curved surfaces and cost effective rapid prototyping of expanded polystyrene foam is carried out. The experimental part of the study falls in two parts. The first presents a number of large-scale cutting samples combining linear cuts with and without inclination measured from the horizontal direction of cutting, while in the second, the thermal phenomena in the process are studied based on infrared measurements of the hot-blade tool made by observation during the cutting process. A novel measurement method for determination of kerfwidth (i.e., the gap space after material removal) applying a commercially available large-scale optical 3D scanning technique was developed and used. A one-dimensional thermo-electro-mechanical numerical model for Hot-Blade Cutting similar to the one previously proposed by Petkov and Hattel (Int J Machine Tools Manuf 107:50–59 2016) for Hot-Wire Cutting of Polystyrene foam is used to simulate the process and describe the effects taking place within the hot-blade during different cutting procedures. The obtained results are graphically presented and discussed in relation to the aim for higher geometrical accuracy of the Hot-Blade Cutting process.

Keywords Hot-blade cutting · Kerfwidth · Expanded polystyrene (EPS) · Thermo-electro-mechanical modelling

Nomenclature

c_p	Specific heat of blade
D_{in}	Internal diameter of blade cross-section
D_{out}	Outer diameter of blade cross-section
D_h	Diameter of EPS holes
E	Young's modulus of blade
F_{kerf}	Kerfwidth as a function of temperature
H_f	Volume specific melting energy of EPS
$k(T)$	Heat conductivity with respect to temperature
L	Length
$L2$	Length of domain 2
N	Number of discretized elements
$\dot{Q}_e'''$	Volumetric specific energy generated
$\dot{Q}_{loss}'''$	Volumetric specific energy loss
Ra	Rayleigh number
S	Speed of cutting
T	Blade temperature
T_{air}	Temperature of air
$T_{initial}$	Initial temperature blade
t	Time
u	Mechanical displacement in x direction
V	Voltage
$V1$	Magnitude of voltage at left boundary
V_m	Volume of blade
x	Position in direction of the length of the blade
$\alpha(T)$	Coefficient of linear thermal expansion-temperature dependent
Δ	Finite difference
ϵ	Emissivity
ρ	Density of blade
$\sigma(T)$	Electrical conductivity with respect to temperature
τ	Normal stress in blade

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1 Introduction

Rapid prototyping has been a topic of interest ever since the 3D computer-aided design (CAD) and large-scale CNC machines were developed in the 1980s. Today, a large number of rapid prototyping techniques are available for scientific and industrial applications, with the most well-known of these being 3D Printing, CNC milling, laser cutting, water-jet cutting (WJC), stereolithography (SLA), selective laser sintering (SLS), and the field is rapidly expanding in various scopes and directions. Particularly interesting are the prototyping processes which involve forming of soft and light-weight materials, e.g., expanded polystyrene (EPS) and extruded polystyrene (XPS), since the prototypes made from these materials are attractive for small- and large-scale designs due to their availability, durability, good strength, and low price.

Relatively few methods are usually applied in terms of forming these materials, e.g., CNC-milling and WJC, but the most popular is the hot-wire cutting (HWC) method, wherein a thin heated wire is used to melt and subsequently form the soft plastic into a desired shape. Different devices and tools for hot-wire cutting are now available on the market as the development in the process in the recent years has been moving toward implementation of CNC and robotic manipulators. Moreover, a number of researchers have been active in investigating the HWC process and its related applications in the past, e.g., [1] proposed an application for HWC for tissue culture flasks, [2] described a method for HWC of optical fibers, and [3] suggested a spring-loaded hot-wire device for polystyrene cutting. A gradual development in the research studies was observed after the 90s, with the works of [4] who proposed a variable lamination HWC method, [5] used a commercial software to calculate the affected melting zone close to the hot-wire, [6] investigated the HWC process for vertically inclined hot-wire cutting, and [7] suggested a simplified way for calculating the power of the hot-wire and the kerfwidth. Moreover, [8] proposed a force-feedback temperature control method correlating applied power, temperature, and measured force, [9] studied the process in details and suggested a number of control strategies for successful hot-wire cutting and [10] measured the volumetric specific energy required for removal of EPS/XPS during HWC. Additionally, [11] proposed an acoustic-feedback control method for HWC, while [12] suggested different correlations for process parameters using the wire linear heat input, wire current, and temperature. The authors of the present work proposed in [13] an in-house numerical model predicting kerfwidth and temperature of the wire during HWC, and [14] investigated the melting of EPS due to standing still of the wire inside the material by implementing well-known fluid correlations for natural convection.

In recent years, the need for more complex large-scale EPS prototypes for commercial and research projects has increased, and this has resulted in the development of a new method for EPS/XPS cutting, i.e., the so-called hot-blade cutting (HBC) in which the cutting hot-wire is replaced by a flexible hot-blade thus allowing for production of geometrically more advanced double-curved prototypes. The HBC method is also considered to be more environmentally friendly and cost-effective in comparison to the conventional CNC-milling, since HBC is significantly faster in speed of fabrication and less energy is required for production of a given pre-defined shape. This is particularly valid for large-scale industrial projects where more than 100 m² of double-curved surface is to be manufactured. The HBC was initially proposed and investigated by [15], while the blade design was optimized in terms of material, shape, and power consumption by [16], and [17] continued the research with respect to the geometrical rationalization of prototypes and prediction of the cold-state-blade shape. The research in HBC was further continued by [18] who developed a bent-blade-tool robotic manipulator for multi-step cutting (similar to milling) of large double curved EPS/XPS prototypes, while [9] modeled the process by using the well-known commercial software ANSYS and proposed a relationship between force, kerfwidth, and effective heat input. In the above mentioned publications, the researchers were successful in building a well working HBC device, though they were challenged in three main areas—material selection of the blade, overall geometrical representation of the final prototypes and keeping fine surface quality. The recently published work of [19] is the first publication to meet these challenges, as large scale double curved EPS prototypes are applied as moulds for concrete casting.

In the current work, a hot-blade cutting process similar to the one proposed by [19] is investigated and studied based on a previously developed thermo-electro-mechanical model for hot-wire cutting proposed by the authors [13].

2 Modelling and governing equations

The thermo-electro-mechanical phenomena that take place in the electrically heated blade, including prediction of the kerfwidth during cutting, is modeled. Since the wire and the blade are geometrically not identical, some important modifications to the original model [13] are made to accommodate for the differences. The overall model structure remains the same, as the model consists of three differential equations, i.e., the one-dimensional heat diffusion equation used for obtaining the temperatures in the blade, the one-dimensional electrical diffusion equation calculating the voltage distribution in the blade and the one-dimensional equilibrium equation including thermal strain for finding the

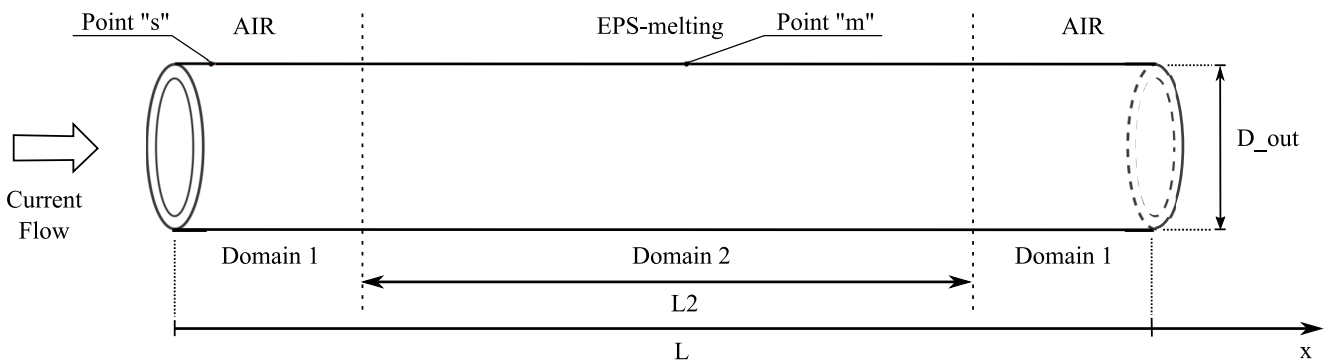


Fig. 1 Geometry of the blade applied in the one dimensional mathematical model

total displacement of the blade due to thermal expansion. In mathematical terms, the equations are described as follows:

$$\rho c_p(T) \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k(T) \frac{\partial T}{\partial x} \right) + \dot{Q}_e''' + \dot{Q}_{loss}''' \quad (1)$$

$$\frac{\partial}{\partial x} \left(\sigma(T) \frac{\partial V}{\partial x} \right) = 0 \quad (2)$$

$$\frac{\partial}{\partial x} \left(E \left(\frac{\partial u}{\partial x} - \alpha(T) \Delta T \right) \right) = 0, \quad (3)$$

where ρ , $T(t, x, V, \dot{Q}_e''', \dot{Q}_{loss}''')$, x , t , $V(t, x, T)$, E , u , ΔT , $\dot{Q}_e'''(t, x, V, T)$ and $\dot{Q}_{loss}'''(t, x, V, T)$ are blade density, blade temperature, position in direction of the length of the blade, time, voltage, Young's modulus, mechanical displacements in x direction, temperature difference taking the initial configuration as reference, volumetric specific energy generated due to electric resistance heating, and volumetric specific energy lost to the surroundings, respectively. The terms $k(T)$, $\sigma(T)$, $c_p(T)$, and $\alpha(T)$ denote heat conductivity, electrical conductivity, specific heat capacity, and coefficient of linear thermal expansion all being dependent on temperature. Temperature, voltage, $\dot{Q}_e'''$ and $\dot{Q}_{loss}'''$ are coupled between the equations. Further information on the individual terms and the computational methods for solving the governing equation can be found in [13]. Moreover, the blade in its transverse direction is in contact with two different environments, i.e., air and EPS, denoted—domain 1 and 2, respectively, as shown in Fig. 1. In both of the domains, the term $\dot{Q}_{loss}'''$ is responsible for the blade's heat loss to the surroundings—air and EPS, and in case of domain 2 the loss of heat due to melting of the EPS at the surface of the blade is described as follows:

$$\dot{Q}_{loss}''' = H_f S F_{kerf}(T) dx/dV_m \quad (\text{Domain 2}), \quad (4)$$

where H_f is volume specific melting energy of EPS foam, S speed of motion of the blade during cutting, $F_{kerf}(T)$ kerfwidth as a function of temperature determined during experimental investigations, dx locally evaluated blade length and dV_m locally evaluated blade volume according to the applied mesh. The expression for $\dot{Q}_{loss}'''$ is evaluated

locally, since temperature, speed and domain's length may vary in the x direction.

The loss of heat in the sections of the blade surrounded by air (domain 1) is governed by the natural convection heat transfer coefficient, which is assumed to follow the correlation of [20] for the range of $Ra < 3$, as additional information on the topic can be found in the works of [21]. The boundary conditions applied for the differential equations ((1), (2) and (3)) are as follows:

$$\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0 \quad \left. \frac{\partial T}{\partial x} \right|_{x=L} = 0 \quad (5)$$

$$V_{x=0} = V1 \quad V_{x=L} = 0 \quad (6)$$

$$u_{x=0} = 0 \quad \tau_{x=L} = 0 \text{ MPa}, \quad (7)$$

where $V1$ is a boundary condition describing an applied voltage of a particular size, while τ^1 denotes applied stress in direction x at $x = L$. The applied stress at the end of the blade is set to zero, allowing only the thermal contribution of the strain (i.e., the thermal expansion) to be computed and stored for later use. In this paper, the mechanical out-of-plane deformation of the blade in 3D is not part of the analysis. On the other hand, the space-dependent thermal strain calculated by the current model plays an important role for determination of the 3D blade shape during thermo-mechanical cutting and can be used for time-optimized computation of the 3D shape. The blade's shape is time and temperature dependent and for this reason the thermal expansion pattern calculated by the model can be a useful supplement to further 3D shape calculations.

The model is discretized according to [13], as the one dimensional space is divided into $N = 500$ elements, where $\Delta x = 0.00296$ m is the length of a single element, and the time step is chosen to be $\Delta t = 0.5$ s in order to ensure stability of the solution. Table 1 presents the main parameters used in the model. Temperature-dependent data for $c_p(T)$

¹Letter τ is used to denote the normal stress in the blade, since the letter σ was already used to indicate another parameter— the electrical conductivity.

Table 1 Table showing model parameters for both blade dimensions and material and expanded polystyrene foam (EPS)

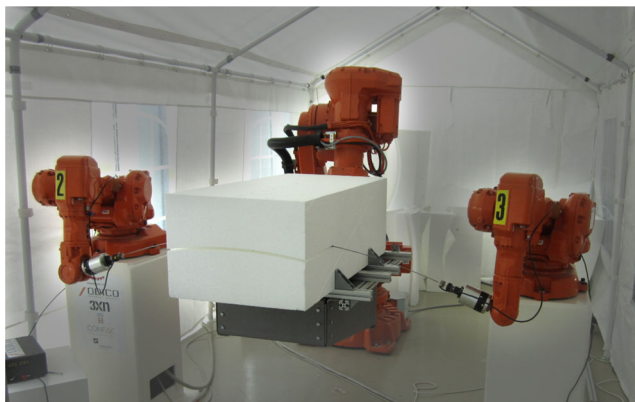
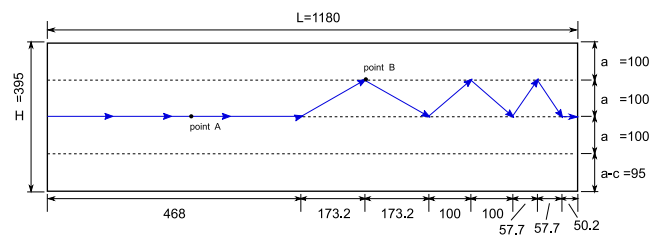
Parameters		
Parameter	Magnitude	Unit
L	1.48	[m]
L_2	0.545 ± 0.002	[m]
D_{out}	3.0	[mm]
D_{in}	2.5	[mm]
ρ	7990	[kg/m ³]
E	193	[GPa]
$T_{air} = T_{initial}$	~ 20	[C]
H_f	8,260,000	[J/m ³]
ϵ	0.94	[–]

The dimensions of the blade geometry- L , L_2 , and D_{out} are presented in Fig. 1 and denote length of the blade subjected to resistance heating, length of domain 2-accounting for the part of the blade in contact with the EPS and outer diameter of the blade, respectively. The value for H_f is experimentally determined

and $k(T)$ are taken from [22], while the electrical conductivity $\sigma(T)$ is based on the works of [23], and the linear temperature expansion coefficient is based on [24].

3 Experimental investigation

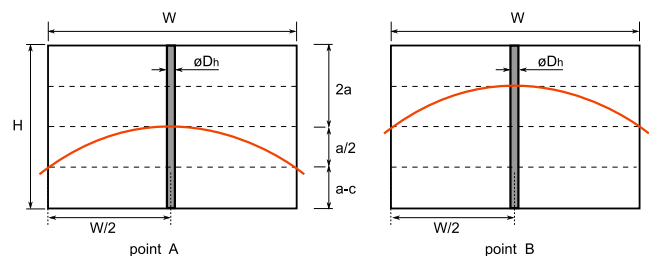
The experimental investigation in this work was performed using an 18-axis, tri-robot system available at the robot-lab of Danish Technological Institute(DTI), Odense. The system consists of two IRB140 robots, manipulating the two ends of the blade, and a single IRB6620 robot holding the the EPS block and positioning the block in the space with

**Fig. 2** Experimental set up for 18-axis, tri-robot system, without IR camera attached**Fig. 3** Diagram of the experimental sample made of EPS foam with dimensions. The arrowed line presents the cutting path

reference to the blade, see Fig. 2. By coordinated motion between the three robots, the cutting tool, i.e., the blade, is carefully brought into contact with the EPS material in order to form it by means of cutting into a desired shape. Due to the high amount of rotational axes of the set-up, a high freedom of double-curved shapes can be achieved, which is beneficial for the final users—designers, architects, and artists who are aiming for prototypes and shapes giving a particular expression in term of aesthetics.

The main cutting tool—the blade—is made from a hal-low circular cross-section tube with dimensions $L = 1.5$ m, $D_{out} = 0.003$ m, and $D_{in} = 0.002$ m, made of a stainless steel alloy. The blade is heated with resistance heating created by a DC current passing through the tool. The power is supplied by a Manson Maas SPS-9400 device connected to the end of the blade by small steel connectors.

The investigation of the process was performed by cutting a number of experimental samples and post-processing the gathered data. The experimental samples were made of EPS blocks, with dimensions $L = 1180$ mm, $H = 395$ mm, and $W = 545$ mm, cut with a hot-wire robotic equipment. The material used for production of the samples was Sun-dolitt EPS 150. The experimental cutting of the samples was performed with HBC technique according to the diagrams in Figs. 3 and 4. Figure 3 presents the cut-path of the blade's tool center point(TCP) with respect to position inside the EPS samples. Each individual experimental sample incorporates seven different cuts representing seven different single

**Fig. 4** Schematic diagram of the experimental sample—front view. Curved lines present the position of the blade during cutting operations, as point A and B represented horizontal movement of the blade as shown in Fig. 3

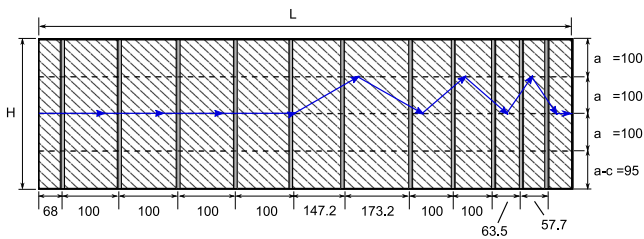


Fig. 5 Cross-section diagram of the experimental sample with hole positions. The *arrowed line* presents the TCP cutting path

experiments, starting with a horizontal cut which is followed by a series of horizontally inclined cuts with angles of $\pm 30^\circ$, $\pm 45^\circ$, and $\pm 60^\circ$. Figure 4 schematically presents the blade position and bending during the cutting operations inside the EPS block. A series of specially pre-cut vertical circular holes of $D_h = 20$ mm were made (prior to the experiments) inside the EPS blocks to allow for a visual observation of the blade during cutting operations. The holes were made with a hot-wire robotic equipment according to the cut-path shown in Fig. 5. Based on an optical monitoring through the holes with a high-end IR camera FLIR SC7600, the temperature measurements during the cutting procedures were made. The temperature measurements were performed with capturing of IR images at rate of 5 Hz made on a small portions of the blade painted with a highly-durable black paint of known emissivity used for accurate temperature reading.

Despite the differences in emissivity between the parts of the blade, i.e., those painted and non-painted with the special paint, no local difference in the final kerfwidth or the final surface quality of the fabricated EPS pieces was observed. Based on these observations, it is assumed that—either the painted and non-painted (but highly oxidized) parts of the blade had similar thermal emissivity thus having equal radiative impact on the EPS, or the radiative fraction of the heat transfer was not dominant in the process, and

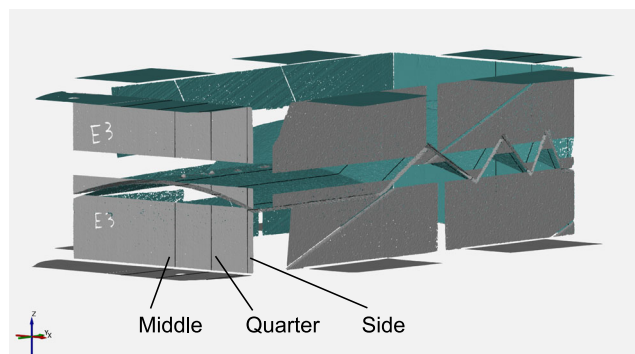


Fig. 6 Virtual assembly of the two scanned parts of an experimental sample after alignment. Three cross-sections (Middle, Quarter, and Side) are made through the length of the assembly for the purpose of kerfwidth measurements at the given cross-sections

subsequently possible differences had a minor effect on the material removal.

The kerfwidth, i.e., the gap in the EPS material left behind the moving-cutting blade-tool, comes as a result of a complex heat transfer process involving both melting and evaporation of EPS foam as well as displacement of the melted and vapor fractions of the material. As previously mentioned in [13], the complex fluid dynamics and motion of the different phases of the EPS foam is a problem beyond the scope of the present work, and for this reason, in the numerical calculation, the kerfwidth is introduced in terms of an empirically-based correlation between the kerfwidth and the temperature. Hence, the physical measurement of the kerfwidth in the experimental samples is an important issue that need to be addressed. In previous publications, e.g., [9, 12] and [13] the kerfwidth was determined by the so-called middle-piece method (MPM), where two subsequent cuts with same process parameters are made in the material and then the kerfwidth is determined by measuring the difference between two specific distances — (1) the constant offset-distance between the two subsequent cuts and (2) the height (or length depending on the direction) of the EPS piece produced in-between the subsequent cuts.² The MPM method is shown to give satisfactory results for the experimental cuttings with hot-wires in the previous works of [9] and [13], but for the particular case of the hot-blade cutting, it is considered unreliable, since the constant offset-distance (prerequisite for applying the MPM) cannot be accurately maintained in the later process, due to the flexibility of the blade, an effect previously confirmed by [16] who stated that the difference in position between two cuts with equal parameters is less than 0.25 mm. The cutting of double curved shapes, requires the blade-tool to be flexible in order to bend into a desired shape while at the same time it needs to sustain the negative effects of the high temperatures to which it is subjected during the cutting. Both of these tool requirements contribute to the increasing of the sensitivity of the tool with respect to unexpected external influences and disturbances (as, e.g., hard/wet spots and drag) thus decreasing the overall geometrical repeatability at the mid-blade cutting range. Moreover, changes in the blade's temperature profile do cause a shape change in the blade's bending profile and thus additional inaccuracies are introduced. To overcome these difficulties and properly measure the kerfwidth in the HBC process, the authors of the present work proposed a new measurement technique based on a Single-Cut(SC) principle, which is used in the experimental part of the work. The SC method is based on preparation of the sample pieces and capturing, aligning and

²For more details regarding the MPM method, the reader may refer to the work of [9] and [13], where the method is applied for hot-wire cutting of EPS.

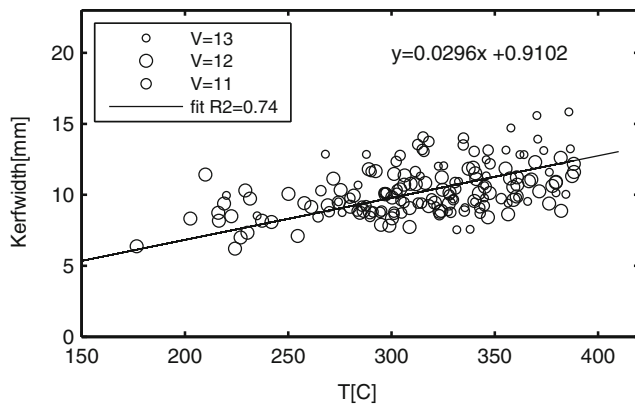


Fig. 7 Kerfwidth with respect to temperature for horizontal hot-blade cutting. The *point size* corresponds to cutting with different applied voltages from 11 to 13 V

observing the two parts of the single-cut-piece resulting at the end of the HBC cutting process. Once the cutting is over, the two separate parts of the sample are brought in a Metrology Lab, where they are scanned with an optical 3D scanner ATOS Triple Scan and subsequently a virtual 3D mesh for each individual piece is made. The two virtual pieces are assembled and aligned using a 3D analyzing software GOM Inspect. Once this step is finished, it is clearly visible on the virtual assembly that there is the missing space between the two aligned parts of the sample, which represents the kerfwidth. Using the flat block geometry sides, the two pieces of the samples were properly aligned. The alignment of the virtual pieces in the direction of the sample height, i.e., normal to the surfaces produced after the single cutting and where the missing part of the block is (that is the kerfwidth), was achieved based on linear physical marks (or scratches) made prior to the hot-blade cutting on the surface of the sample blocks. Based on the marks, one can accurately match the pieces and bring them to the initial high of the sample. A drawback of the method is the uncertainty in

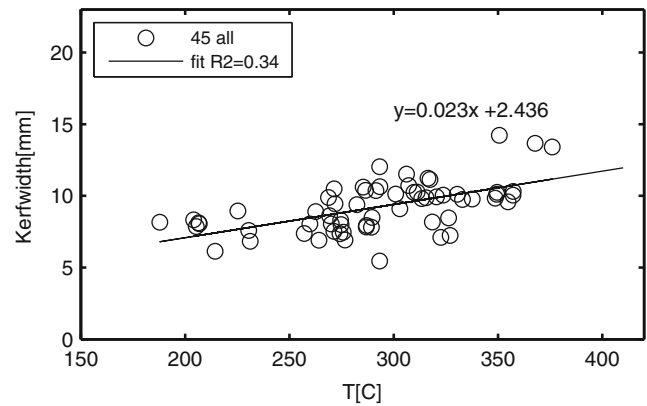


Fig. 9 Kerfwidth with respect to temperature for hot-blade cutting of EPS with 45° inclination from the *horizontal*. The *points* represent cutting with +45° and −45° inclination

the alignment procedure, which is often sensitive and prone to slight angular errors in a magnitude below 0.1°, which negatively influencing the overall kerfwidth measurements with errors estimated being within ± 1.5 mm. The kerfwidth measurement on the virtual assembly was made by creating a virtual cross-section, as shown in Fig. 6, in the alignment and exporting the fine geometrical mesh points of the same to MatLab. The horizontal cuts of the kerfwidth are determined by the perpendicular vertical distance between the two aligned sample pieces, while for the cuts with inclination with respect to the horizontal plane, the kerfwidth is derived from the normal to the surface distance between the aligned sample pieces. The mid-block TCP kerfwidth is defined at 240 mm (to avoid intersection with the observation holes) from the surface in the dimension of the Width, and is evaluated for each of the 11 circular holes.

Based on the kerfwidth measurements, comparative analyses were made with other process parameters. Figure 7 shows the comparison between kerfwidth and temperature for the experimental samples made for horizontal HBC

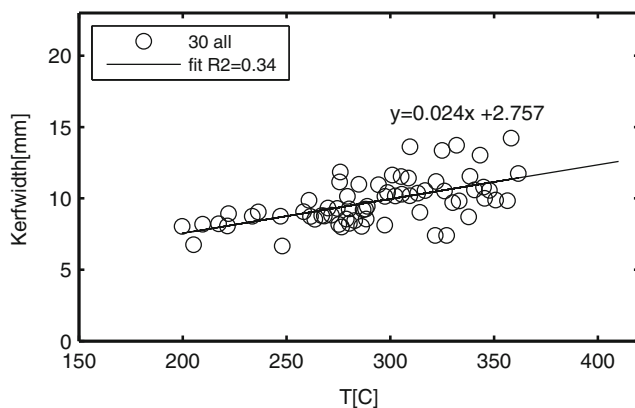


Fig. 8 Kerfwidth with respect to temperature for hot-blade cutting of EPS with 30° inclination from the *horizontal*. The *points* represent cutting with +30° and −30° inclination

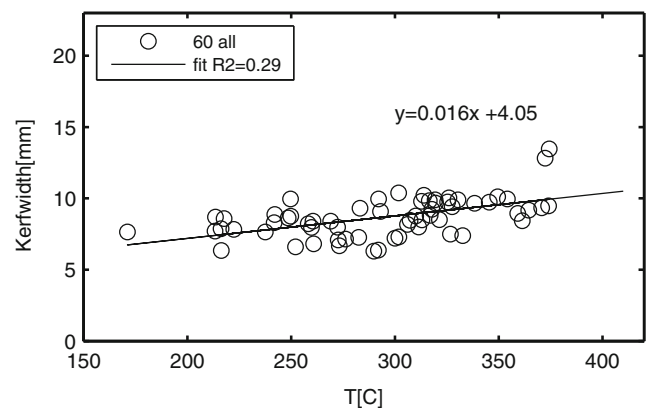


Fig. 10 Kerfwidth with respect to temperature for hot-blade cutting of EPS with 60° inclination from the *horizontal*. The *points* represent cutting with +60° and −60° inclination

cutting. Figures 8, 9, and 10 show the relation between kerfwidth and temperature obtained for cutting with angles of 30° , 45° , and 60° , respectively.

In all the relations between kerfwidth and temperature (Figs. 7, 8, 9, and 10), there is a deviation of the data points with respect to the approximated linear fit, as the deviation is expressed in a higher degree in the figures presenting results for experiments with inclined cutting conditions. The deviation is probably caused by alignment errors and errors in defining the accurate cutting temperatures, the latter being estimated to fall within $[0\text{--}50]^\circ\text{C}$. The errors in defining the temperature of cutting are caused by the uncertainty of defining the first temperature reading point during the camera measurement of the blade inside the circular holes. During the cutting process, much of the EPS around the blade is melted and evaporated, which causes a large amount of vapor to be present in close vicinity to the blade. During the time instances when the blade enters into the circular holes, some of the vapor enters in the circular hole

prior to the blade thus causing the first reading point to be difficult to be determined and in some cases to be incorrectly measured. This problem is particularly pronounced for the holes at the inclined cutting experiments, where in some cases the holes were full of vapor—observed as smoke. Figure 11 shows the IR camera readings with and without smoke present in the holes. The temperature scale on the figures is shifted, since the paint emissivity coefficient and the camera calibration data are not included in the figures, but are accounted for in the post-processing of the data.

The best fit between kerfwidth and temperature is achieved for the horizontal HBC cutting, where the data fit accounts for 74% of the measured points, as shown in Fig. 7. The applied voltage during cutting does not seem to influence the trend of the data points in the kerfwidth-temperature comparison. The kerfwidth-temperature graphs for cutting angles of $\pm 30^\circ$, $\pm 45^\circ$, and $\pm 60^\circ$ are capable of accounting for approximately a third part of the points—with highest R^2 of 0.34 for the cases of cutting with $\pm 30^\circ$ and

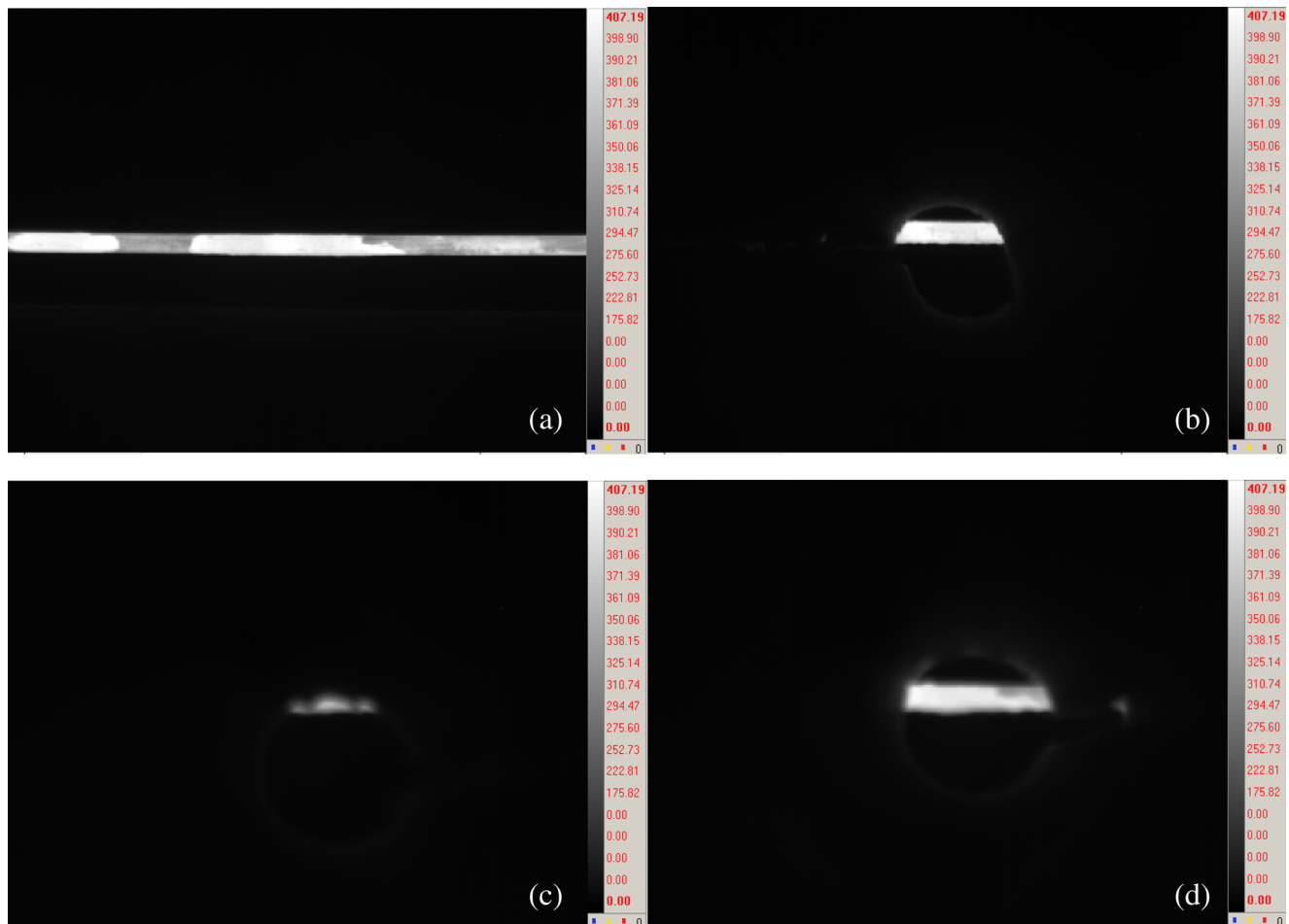


Fig. 11 Blade temperature measurements during cutting process inside the circular holes. **a** IR temperature measurement—blade in air, i.e., outside EPS. *Bright spots* are the painted parts of the blade. **b** IR temperature measurement—clear view of the blade in the circular

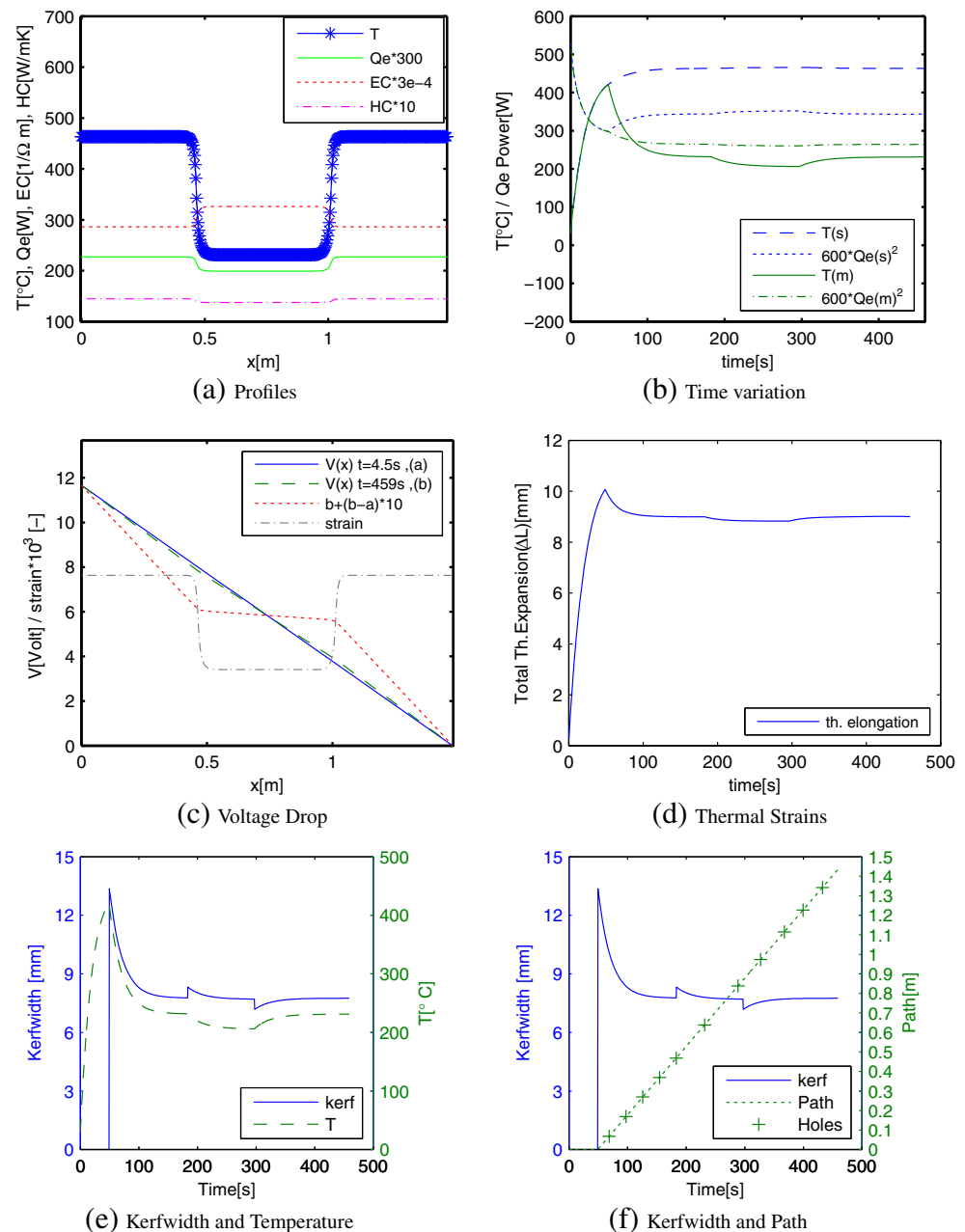
hole. **c** IR temperature measurement—blade enters in the circular hole. Image blurred by smoke in the hole. **d** IR temperature measurement—blade in the circular hole. Image blurred by smoke in the hole

$\pm 45^\circ$. It is still unclear what causes the dispersion of the data with respect to the fitting lines, as probable explanation could be found in the measurement errors and the errors of unpredictable nature as wet and hard spots present in the material. Additionally, the HBC method produces a higher kerfwidth in comparison to the hot-wire method, the kerfwidth being usually between 5 and 16 mm, which results in higher amount of melted/evaporated EPS material and thus making the local material defects to play a bigger role for the single measurement point. Despite the scatter of the data, the fitting lines represent well the overall trend of the relation between the kerfwidth and the temperature, and for this reason are included in the numerical simulation.

4 Results

The results of the numerical model for hot-blade cutting are presented in Fig. 12. The model outputs different internal process parameters, which are shown with respect to space and time. The plots in Fig. 12 present a simulation for a single HBC cutting of EPS according to the experimental cut-path presented in Figs. 3, 4, and 5. Each of the simulations and experiments, performed in the current work, starts with initial pre-heating of the blade in air until a cutting temperature is reached, which is followed by initiation of robotic motion to bring the blade in contact with the EPS material and through progressing inside the EPS material and

Fig. 12 The figures show simulated temperatures and various parameters with respect to space and time for the cutting process. Figure 12a shows temperature (T), generated energy through resistance heating (Qe), electrical conductivity (EC), and thermal conductivity (HC) with respect to position x (space) at $t \approx 459$ s. Figure 12b shows temperature (T) and Qe with respect to time for two points “s” and “m” as shown in Fig. 1. Figure 12c shows the voltage drop through the blade with respect to x (space) at two time instants. The difference between the profiles at the time instants is given as a function of $b+(b-a)*10$. Additionally, the local thermal strain of the blade with respect to x (space) is plotted. Figure 12d shows the total thermal expansion of the blade with respect to time for the presented simulation. Figure 12e shows the kerfwidth and temperature variation during cutting of one experimental sample as shown in Fig. 3. Figure 12f shows kerfwidth, passed distance of cut-path and holes’ positions. It should be noted that the cut-path is longer than the sample length(L) due to cutting with inclination from the horizontal direction. Process parameters $V = 11.68$, speed = 3.5 mm/s



maintaining a constant speed of motion, the cutting of the material is done. The same behavior is reflected in Fig. 12, as Fig. 12b shows the time-dependent temperature and the generated energy ($\dot{Q}_e''$) due to electrical heating at two points on the blade—Point “s”(side) and Point “m”(middle), as the position of the two points is given in Fig. 1. The two points mentioned (“s” and “m”) present the behavior of the blade in the two surrounding environments in which it is operating during the process of cutting—air(s) and EPS(m). Since the domains affect the cooling of the blade in different ways, a temperature difference is introduced in the blade as soon as the cutting of EPS is initiated at $t = 50$ s. As the melting and evaporation of EPS is exerting higher local cooling on the blade, the temperature at those points is lower in comparison to the parts of the blade in air. Moreover, the resistance heating is influenced by the local temperature variations and the heat generated through resistance heating is locally changing during the cutting. The additional variations in both parameters after $t = 170$ s are caused by the initiation of cutting with angles according to Fig. 3. The temperature and the generated energy ($\dot{Q}_e''$)—with respect to space(x) on the blade are presented in Fig. 12a together with space dependent electrical conductivity and heat conductivity of the blade. Figure 12c shows the space-dependent voltage drop of the blade. The sections of the blade exposed to higher temperature have consequently a higher electrical resistivity and as a result a higher voltage drop. Since the stainless steel alloy experiences much change of electrical resistivity with respect to temperature, a relatively large change in voltage drop is observed between the domains 1 and 2.

Parallel to the voltage is presented the local thermal strain of the blade with respect to position. Figure 12d shows the total thermal elongation of the blade with respect

to simulated time during cutting. The thermal elongation is calculated for a single time-step with the formula $\sum_{i=1}^{500} \alpha(T_i) \Delta T_i \Delta x_i$, where i is the index describing the space discretization. Based on the total blade length, calculated during heating, and cutting operations, the bended profile of the blade in its hot-state can be found, which is a prerequisite for cutting highly accurate curved shapes. The simulated kerfwidth during cutting is presented in Fig. 12e, f, and is compared to the mid-blade cutting temperature and cut path in direction of the block sample length, respectively. The slight jumps in the kerfwidth at $t = 183$ s and $t = 279$ s are caused by the sudden shift of correlation for cutting of $\pm 30^\circ$ and $\pm 45^\circ$, while the shift for $\pm 60^\circ$ correlation produces an almost insignificant jump in kerfwidth at $t = 378$ s. Moreover, the jumps in kerfwidth are followed by a short transition zone leading to a local steady state kerfwidth for each individual angle correlation. On the other hand, at the instances of sudden kerfwidth change, the temperature is smoothly transitioning between the correlations to account for the different cutting conditions, as shown in Fig. 12e. The cut-path and the holes position for visual observation are presented in Fig. 12f as the cut-path is longer than the block length due to the angled sections of the individual experiments.

A comparison between the experimental and simulated results is presented graphically in Figs. 13, 14, and 15. The results represent a set of 49 individual experimental sequences according to Fig. 3, out of which 31 were successful and are presented below—providing approximately 341 measurement points. Figure 13a–c shows the simulated and experimental kerfwidth for horizontal cutting with Hot-Blade for three different cross-sections of the experimental blocks—at distances 20 mm (Side), 130 mm (Quarter), and 240 mm (Middle) measured from the side of the samples’

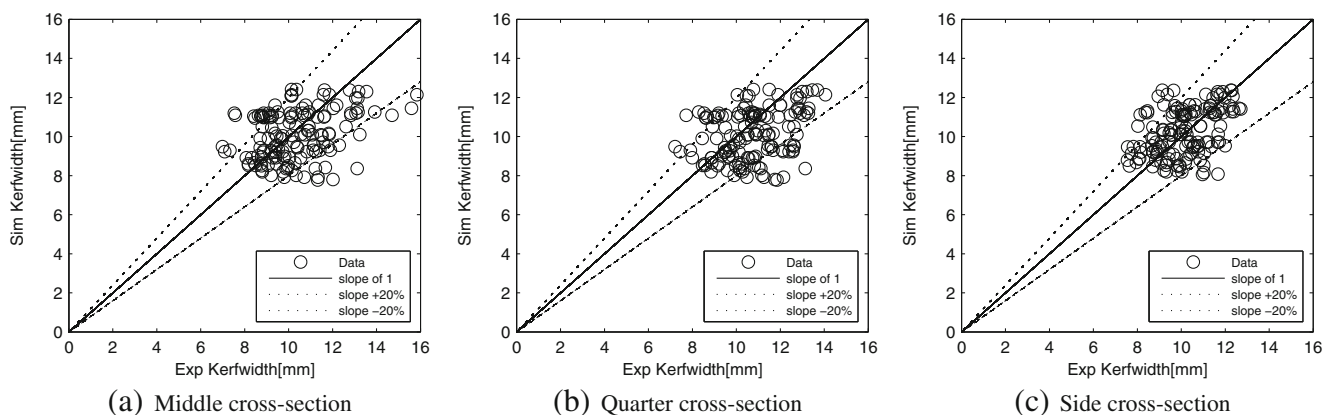


Fig. 13 The figures show the comparison in simulated and experimental results for kerfwidth during horizontal hot-blade cutting of EPS foam. Each data point compares one experimental and one simulated measurement on the surface of the post-cut EPS samples. The points lying on the line of slope 1 correspond to 100% agreement between

the simulated and experimental results, as lines of $\pm 20\%$ error are provided. The graphs show the kerfwidth results for three different cross-sections at the length of the block samples—the Middle, the Quarter, and the Side cross-sections defined at distances 240, 130, and 20 mm, respectively. The cross sections are shown on Fig. 6

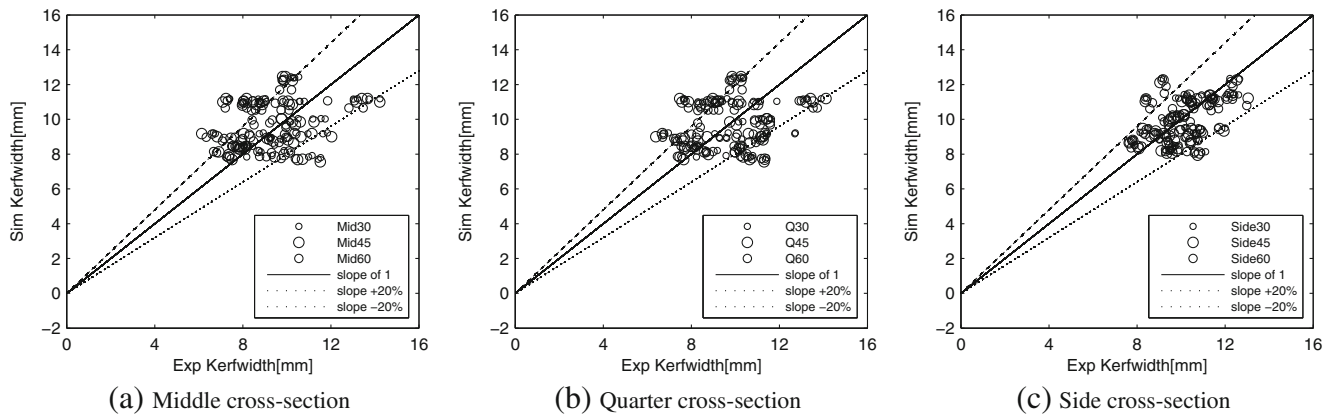


Fig. 14 The figures show the comparison in simulated and experimental results for kerfwidth during hot-blade cutting of EPS foam. Each data point compares one experimental and one simulated measurement on the surface of the post-cut EPS samples. The points lying on the line of slope 1 correspond to 100% agreement between the simulated and

experimental results, as lines of $\pm 20\%$ error are provided. The graphs show the kerfwidth results for different angles of cutting for three different cross-sections at the length of the block samples—the Middle, the Quarter, and the Side cross-sections defined at distances 240, 130, and 20 mm, respectively. The cross sections are shown on Fig. 6

blocks. On Figs. 13, 14, and 15 the points are compared with -respect to a slope of 1, corresponding to a 100% match between the experimental and simulated results. Additional lines for plus and minus 20% errors are provided for comparison reasons. The results presented by Fig. 13a–c show that most of the simulation results are falling within the 20% error margin compared to the experimental kerfwidth measurements. On the other hand, the agreement between the experimental and simulated data is improving in the given order of cross-section frames—Middle (Fig. 13a), Quarter (Fig. 13b), Side (Fig. 13c), as the kerfwidth measurements at the Side show the best match between experimental and simulated results, with a limited number of points falling

outside of the 20% margin. A similar trend is observed for the Fig. 14a–c comparing the simulated and experimental results for cutting with an angle with respect to the horizontal cutting direction. In these figures, a larger number of the compared points fall outside of the 20% error margin, meaning that there is a larger discrepancy between the modeling and experimental work. However, it should be remembered that the simulations performed in the numerical simulation were made based on the previously proposed fits between kerfwidth and temperature (Figs. 7, 8, 9, and 10) which resemble approximately a third part of the given experimental measurements, thus introducing inaccuracy in the model. The best agreement between the

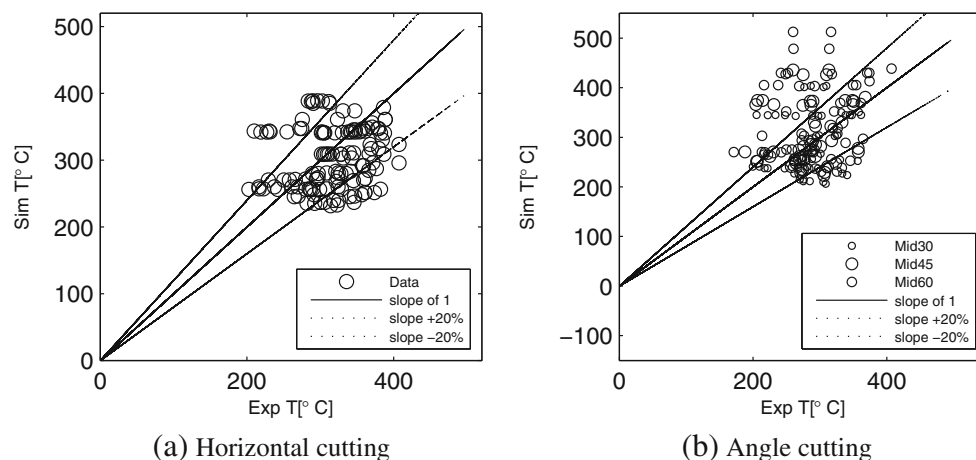


Fig. 15 The figures show the comparison in simulated and experimental results for temperature during hot-blade cutting of EPS foam. Each data point compares one experimental and one simulated measurement on the surface of the blade. The points lying on the line of slope 1 correspond to 100% agreement between the simulated and experimental

results, as lines of $\pm 20\%$ error are provided. All points are measured/taken at middle of the blade cutting tool. Figure 15a shows the temperature comparisons for horizontal cutting, while Fig. 15b for cutting with angles with respect to the horizontal plane

experimental and simulated results for the HBC with inclination is observed again for the measurements taken from the Side of the EPS block. A possible explanation for the good agreement between the results close to the Side of the EPS block for horizontal and inclined cuts is based on the fact that the alignment on the experimental samples was made at the side, and for this reason the points close to the side suffer less from angular errors. Figure 15a, -b compare the experimental and simulated results for the temperature at the Middle cross-section of the samples. The data shows a good agreement between the experimental and simulated results, though a given number of data points falls outside of the 20% margin. Moreover, the results for horizontal cutting (Fig. 15a) show a better agreement in comparison to those for angle cutting (Fig. 15b). The reasons for the mismatches in the result comparisons for temperature could be diverse, as the main sources of errors are possibly—the error in experimental temperature acquisition, the uncertainty in the measured volume specific melting energy of the EPS foam (the latter being with a standard deviation of 15%) and the uncertainty in the kerfwidth-temperature relations. A larger number of experimental data points could cast additional light on the subject and help improving the agreement between the model and the experimental work; however, this is a topic for future work and investigation.

5 Conclusions

Experimental and numerical work for the newly developed process of hot-blade cutting is presented. The process is considered an environmental friendly alternative to the more traditional CNC-milling due to its higher productivity and less overall consumption of energy. The work deals with some of the challenges during investigation of the main parameters of the cutting process, and a new method for measurement of kerfwidth, particularly suited for the Hot-Blade Cutting process, is proposed. The numerical part of the study presented a new model for simulating the effect of internal process parameters such as temperature, generated energy for electrical heating, voltage drop distribution in the blade, local thermal expansion, and kerfwidth which all are estimated based on empirical relations and experimental study performed during the course of this work. The major part of the simulated results for kerfwidth at the side of the investigated samples are found to be within a 20% error margin as compared to the experimental data. Additional research is suggested in order to be able to improve the main empirical relations on which the model is based and increase the overall accuracy of the model.

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